
Generalization in Reinforcement Learning

ActorDQN vs DQN

Xiaojie Gu

University of Wuerzburg

First Advisor: Mr. Sebastian Griesbach | Supervisor: Prof. Dr. Carlo D'Eramo

This presentation was made with assistance from GPT-6 Sol under the guidance of my hand draft.

Outline

1. Introduction
2. Environment design
3. Model Design
4. Experimental results

Introduction: generalization in robotics

Training condition



Table height: 1.0 m Garment: T-shirt
Neat fold in a familiar setting

Unseen test condition



Table height: 1.2 m Garment: jacket
Unsuccessful fold after the change

Success in training does not guarantee success when conditions change.

Images generated by GPT.

Generalization in robotics: Unitree video

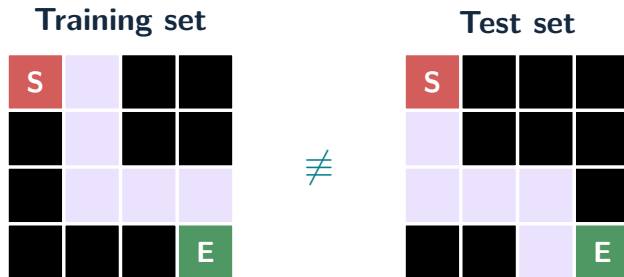


A real-world robotics example

Environment Design

1. Regulation
2. Procedural maze

Regulation: no shared optimal solution

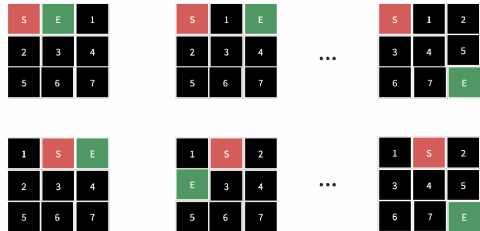


$$\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$$

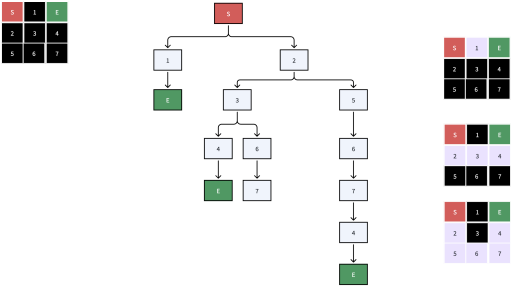
Each original maze and its variations stay in one split; the intended shortest path does not cross splits.

Procedural Maze

1. Enumerate start-goal pairs



2. Generate valid paths



Model Design

1. Idea: classification may be easier than regression
2. DQN
3. ActorDQN

Model Design: Why classification?

Hypothesis: When the goal is to choose one option, learning the choice may be easier than fitting a precise score for every option.

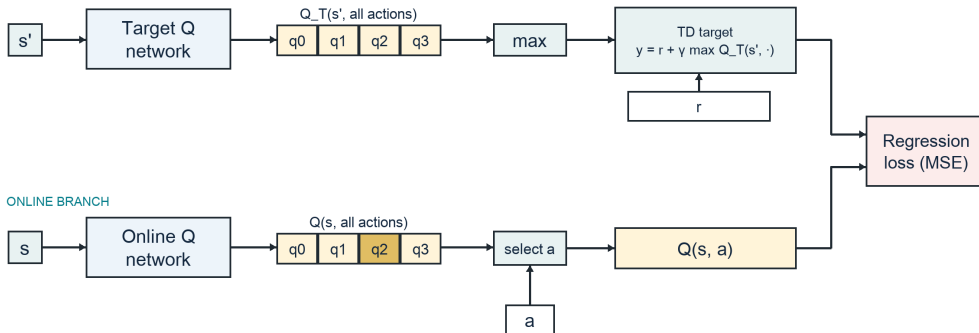
	Book 1	Book 2	Book 3	Book 4	
	0.1	0.2	0.4	0.9	1. score all four
Regression (two steps)	×	×	×	✓	2. choose the best
Classification	×	×	×	✓	learn the best-book label directly

In the maze: books $\leftrightarrow$ actions; scores $\leftrightarrow$ Q-values; selected book $\leftrightarrow$ action label.

Model Design: DQN

REPLAY TRANSITION (s, a, r, s')

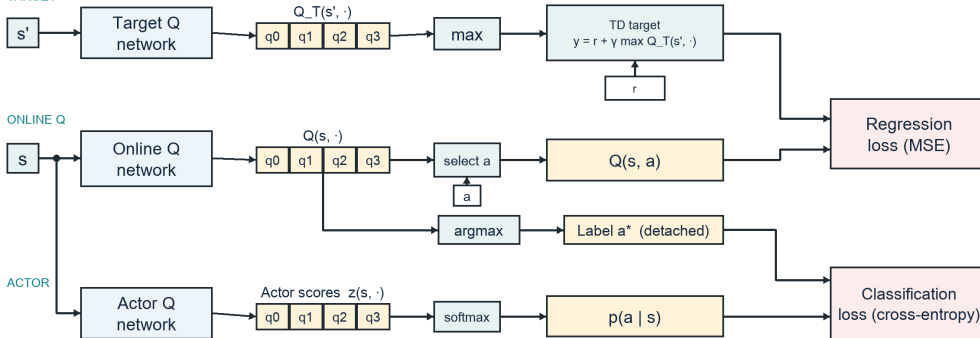
TARGET BRANCH



Model Design: ActorDQN

REPLAY TRANSITION (s, a, r, s')

TARGET



Experimental Results

1. Performance in mazes

Full held-out sets for 4×4 and 5×5 mazes.

2. Justification for the idea

Performance in Maze: 4×4

Held-out set: 201 mazes per seed Step limit: 160		
Policy	Success rate	Mean steps, successful mazes
ActorDQN	0.828 ± 0.029	25.95 ± 3.65
Basic DQN	0.111 ± 0.072	4.42 ± 0.94
Uniform Random	0.740 ± 0.048	63.86 ± 4.04

Mean $\pm$ sample SD across three seeds.

Performance in Maze: 5×5

Held-out set: 2,973 mazes per seed Step limit: 250		
Policy	Success rate	Mean steps, successful mazes
ActorDQN	0.673 ± 0.027	48.26 ± 5.09
Basic DQN	0.129 ± 0.036	8.33 ± 0.45
Uniform Random	0.564 ± 0.004	122.25 ± 1.22

Mean $\pm$ sample SD across three seeds.

Generalization Gap

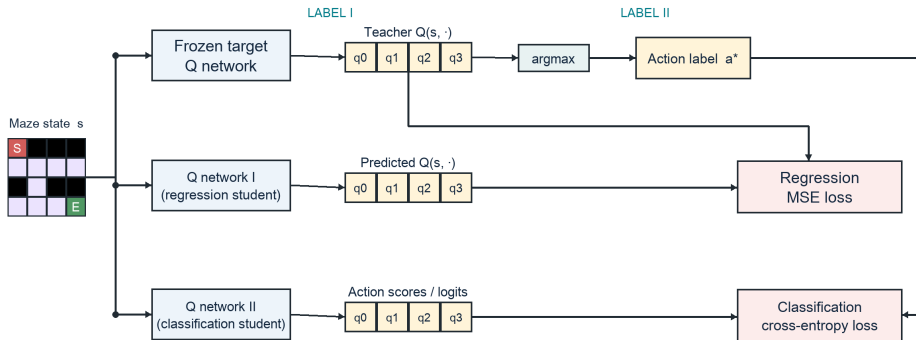
Gap = training-set mean – held-out mean;

Maze	Policy	Δ steps, all	Δ steps, success	Δ success rate
4×4	ActorDQN	-6.07	-2.92	+0.026
	Basic DQN	-15.72	+0.12	+0.102
5×5	ActorDQN	-3.89	-1.32	+0.014
	Basic DQN	-1.83	+0.03	+0.008

A smaller absolute gap is better.

Justification for the Idea

Two frozen teachers are tested separately: DQN target and ActorDQN target.



Matched students share architecture, initialization, minibatches, and update budget.

Classification vs. Q-value Regression

Teacher	Objective	Agreement 10k updates	Agreement 102.4k updates	Maze success 102.4k updates
DQN	Classification	0.868	0.899	137/201 (0.682)
	Regression	0.800	0.853	116/201 (0.577)
ActorDQN	Classification	0.964	0.967	175/201 (0.871)
	Regression	0.910	0.955	162/201 (0.806)

Action agreement means matching the teacher's greedy action on held-out observations.

Thanks